

Assignment-Series01-Rstudio

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1. Conclusion in 4.1

Running a Mann–Whitney U test on the 22-student dataset (7 females, 15 males) gave a p-value around 0.003–0.005 and a large effect size ($r = 0.60$, confirmed by `wilcox_effsize()`). This matches the conclusion given in the assignment: there is a statistically significant difference between male and female median weight in this sample, with a large effect size. My result is the same as the one given, which makes sense we used the same non-parametric test appropriate for this small, unevenly-sized, non-normally-distributed sample, applied to the same underlying data.

2. Three most important things I learned

1. How to move data from a raw external file into a clean, typed R object, and how to notice type-parsing issues (like a comma used as a decimal separator turning a numeric column into text).
2. How to build and reshape summary tables with the tidyverse “grammar” (`tibble()`, `group_by()`, `summarise()`), and how to choose and build the right plot (boxplot vs. histogram vs. density) for a given question with `ggplot2`.
3. How to move from a purely descriptive summary to an actual scientific comparison — choosing an appropriate statistical test given the sample size and shape of the data, and pairing it with the right visualization to communicate the result.